Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Advanced grooming solutions validated by canine in-vitro reporter skin models for sensitization*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		***Bio・Medical***	
	Medium Category	Small Category	Medium Category	Small Category
	Industrial Biotechnology	*Bio-Cosmetics / Materials, Other Industrial Biotechnology*	*Medical Bio*	*Other (veterinarian) Biopharmaceuticals / Biomaterials*
Concept of Proposed Technology	*We propose an in-vitro dog skin reporter model to detect and evaluate sensitization effects of care and grooming ingredients and products. These reconstructed epidermis model mimics in vivo conditions with an intact barrier and use immortalized canine keratinocytes for high reproducibility. The Nrf2 Sensitization pathway is monitored via a stably integrated inducible reporter protein, enabling fast and reliable assessment of harmful or protective effects on dog skin. This enables identification of both damaging and protective effects and allows cruelty-free evaluation of ingredients for care and grooming products (spray, shampoo, cream). The model also enables testing hydrophobic substances or mixtures for the first time.*			

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Rationale for Joint R&D Project	*Dogs, the most prevalent companion animal, have highly sensitive skin due to their thin epidermis and weak keratinization, making them prone to dermatitis and allergies. As a result, the demand for specialized, safe, and effective skin-care products continues to grow [1].* *Against this background, it is essential to consider how such products are regulated. In the EU, pet shampoos and grooming products are generally regulated as detergents. Accordingly, manufacturers must comply with the Detergents Regulation (EC) No. 648/2004 [2] - covering surfactant biodegradability, compositional limits, safety evaluations based on existing toxicological data, and detailed labelling - together with national market-surveillance checks; there is no formal pre-market approval [3]. By contrast, South Korea likewise has no pet-shampoo-specific pre-market approval scheme. Instead, manufacturers typically rely on existing chemical and toxicological data under general consumer-product rules [4], and the country's broader move toward animal-free science further encourages non-animal safety assessments [5].* *Crucially, neither framework requires proof of compatibility with canine skin. This is notable because canine skin differs substantially from human skin—most visibly in its thinner stratum corneum and epidermis. When the stratum corneum is thin or compromised, substances can penetrate more rapidly into deeper layers, be absorbed in higher amounts, and, in some cases, enter the bloodstream more readily. As a result, both beneficial and adverse effects can be amplified.* *Therefore, even when ingredients and finished formulations are considered safe under human-skin standards, such findings do not automatically translate to dogs. Differences in baseline skin physiology - especially the distinct skin pH and thinner cornified layer - can significantly alter penetration, irritation potential, and overall tolerability. Consequently, dedicated grooming products should ideally be tested specifically for dogs, for example by using species-specific in-vitro canine skin models.* *However, there is currently a lack of reproducible, cruelty-free in-vitro test systems tailored to canine skin. Such models are essential for developing and validating innovative, skin-friendly grooming products. Fraunhofer IGB offers extensive expertise in advanced in-vitro skin models and reporter systems [6-10], but international collaboration is needed to complement this with veterinary, market, ingredient and formulation know-how from a Korean partner. This collaborative effort will establish the first reliable canine in-vitro reporter skin model for sensitization, enabling efficient, species-specific testing for the growing pet care market in Korea and beyond. It will support the development and reliable assessment of non-sensitizing care and grooming products for dogs.*
Key Words	*1. Canine in-vitro reporter skin model to detect sensitizing substances and compounds* *– In this species-specific, reproducible dog skin in-vitro test system, an inducible reporter gene is stably integrated into the epidermal keratinocytes which enables the detection of the activation of the specific Nrf2-sensitization pathway enabling straightforward detection of sensitizing effects on canine skin* *– Monitoring of cellular stress responses (e.g. sensitization Nrf2 pathway) using integrated reporter proteins, which are secreted into the culture medium for straightforward measurement by an easy*

	quick and quantitative read-out. *2. Cruelty-free testing* *– No animal testing required; enables ethical product development; Both cultivation and testing are carried out without any animal-derived supplements* *3. Medium-throughput screening* *– Efficient evaluation of multiple ingredients and formulations* *4. Assessment of skin-damaging and skin-protecting effects* *– Identification and validation of both harmful and beneficial effects for new pet care products*

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	1. *Canine in-vitro reporter skin model for sensitization:* *Development of the first reproducible, species-specific in-vitro skin model for dogs, enabling the detection of sensitizing effects of substances via integrated reporter systems (Nrf2 pathway). These models are built from immortalized specific canine reporter keratinocytes.* 2. *Cruelty-free, medium-throughput testing platform:* *Establishment of an ethical, animal-free test system for the rapid and reliable screening of ingredients and formulations for dog grooming products.* 3. *Assessment of skin-damaging and skin-protecting effects:* *Technology to specifically identify and evaluate both harmful and beneficial effects of substances on canine skin, supporting the development of safe and effective species-specific pet care products.* 4. *Testing of hydrophobic substances:* *Enabling, for the first time, the evaluation of hydrophobic (water-insoluble) ingredients and mixtures in a physiologically relevant in-vitro dog skin model.*			
R&D Target Specifications	*1.Generation of an immortalized canine keratinocyte cell line with stable integrated inducible reporter gene (e.g., Nrf2 pathway) and establishment of a canine in-vitro reporter skin model based on this cell line:* *Model must demonstrate high donor-independence and batch-to-batch reproducibility.* *2. Sensitivity and specificity for sensitization detection:* *The model should reliably detect known sensitizers and non-sensitizers.* *3. Quantitative readout:* *Quantification of the Reporter protein release in the culture medium (e.g., via photometric assay), with a detection limit in the low nanogram per milliliter (ng/mL) range.* *4. Testing of hydrophobic substances:* *The model must enable the application and reliable assessment of hydrophobic ingredients (e.g., creams, sprays, shampoos).* *5. Product efficacy and safety assessment:* *Ability to simultaneously differentiate skin-damaging and skin-protecting effects in a single measurement, using quantitative endpoints for cell viability, barrier integrity, and reporter activation.*			

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R&D Plan	*Research Approach: In the project, the Fraunhofer IGB will develop a reproducible, cruelty-free in-vitro canine reporter skin model for the detection of sensitizing substances.* *This is based on preliminary work on immortalized primary keratinocytes and an established in-vitro dog skin model representing healthy skin expressing important expression markers as native skin. In the proposed project the first in-vitro canine sensitization model will be established. The project approach combines advanced cell engineering, molecular biology, and formulation science. Using this model, potential ingredients for dog grooming products are screened for their sensitizing effects and their impact on the skin barrier, allowing suitable candidates for formulation development to be identified. Finally, prototype products such as shampoos, creams, and sprays will be developed and evaluated using the in-vitro reporter skin model, enabling quantitative risk assessment of sensitization potential.* *Key Tasks & Methodologies:* ***Pre-Study (6 months)*** *Generation of reporter cell line for sensitizing effects (Nrf2 pathway) based on the immortalized canine keratinocyte cell line of IGB* - *Design and cloning of reporter constructs plasmid (e.g., Nrf2 pathway)* - *Stable transduction of the reporter construct into the immortalized canine keratinocyte cell line and selection of a final monoclonal reporter cell line for the detection of sensitizers* *Establishment of 2D screening assays with the canine keratinocyte reporter cell line* - *Characterization of reporter cell line e.g. for specific reporter activity* - *Establishment of 2D screening assay based on the canine keratinocyte reporter cell line including suitable positive and negative reference substances* - *First screening of water-soluble substances* ***Main R&D study (30 months)*** *Establishment of 3D in-vitro skin model with the canine reporter keratinocyte cell line:* - *Construction of skin equivalents using immortalized canine reporter skin cells (see above)* - *Validation of skin barrier function* - *Development of a Standard Operation Procedure (SOP) for establishment and handling of the in-vitro reporter model including quantitative test protocols for sensitization* *Development and screening of substances, compounds or mixtures* - *Selection of a suitable library of candidates* - *Physicochemical and biological characterization (cytotoxicity)* - *Screening of non-/sensitizing ingredients using the reporter skin model*

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	Formulation and testing of grooming products: - *Development of prototype products (shampoo, cream, spray)* - *Application and efficacy testing on the in-vitro reporter skin model* - *Quantitative analysis of sensitization* *Regulatory preparation and technology transfer:* - *Documentation for regulatory approval* - *Preparation for commercialization and licensing* *Expected Milestones & Timeline:* *Month 6: Stable canine keratinocyte reporter cell line for the detection of sensitizers,* *Month 15: Validated canine in-vitro skin reporter model for skin sensitization* *Month 18: At least one identified ingredient with non-sensitizing, skin-friendly properties* *Month 24: First formulation of a skin-friendly grooming product available* *Month 30: Final product formulation tested and evaluated using the model*

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Technical Contributions:* *Fraunhofer IGB is developing the first canine in-vitro reporter skin model for sensitization, based on preliminary work with immortalized primary keratinocytes and fibroblasts and an established basic full-thickness dog skin model representing healthy skin. The work includes generating and genetically modifying immortalized primary canine keratinocytes, integrating reporter constructs (e.g., Nrf2 pathway), and establishing a skin model with an intact barrier function.* *Research Scope: The institute will focus on the design, development, and validation of the in-vitro model, including the creation of standard operating procedures (SOPs), and the establishment of quantitative readouts for sensitization and skin barrier integrity.* *Resources Provided: Fraunhofer IGB provides state-of-the-art molecular and cell biology laboratories (biosafety levels 1 and 2), advanced histology and imaging infrastructure, and experienced personnel with over 20 years of expertise in cell and tissue engineering.* *Specific Tasks:* - *Generation and characterization of stable canine reporter keratinocyte cell line for Nrf2 pathway activation* - *Construction and validation of the 3D in-vitro skin model based on the canine reporter keratinocyte cell line for Nrf2 pathway activation* - *Development of protocols for the application and testing of various formulations (e.g., creams, sprays, shampoos)* - *Quantitative analysis of a specific reporter expression and secetion and barrier function* - *Evaluation of the final grooming product using the established model* - *Preparation of regulatory documentation and support for technology transfer* *Fraunhofer IGB will also actively disseminate results through scientific publications and conferences and will provide access to the developed model for further research or commercial use via licensing agreements including training in the construction of skin models and their handling.*
Expected R&R of Korean SME Partner	*Technical Contributions: The Korean SME is expected to contribute expertise in the formulation and development of innovative pet grooming products, particularly using skin-friendly ingredients suitable for dogs.* *Development Tasks: The partner will be responsible for selecting, sourcing, and formulating candidate ingredients, as well as developing prototype grooming products (e.g., shampoos, creams, sprays) based on the results from the in-vitro skin model.* *Testing and Validation Activities: The SME will collaborate in the testing and validation of ingredients and final products using the canine in-vitro reporter skin model developed by Fraunhofer IGB. This includes supporting the adaptation of test protocols to market needs, particularly by contributing scientific expertise to refine assay conditions, optimize readout*

	parameters, adjust test matrices for different product categories, and ensure the methods align with regulatory expectations, customer requirements, and emerging industry standards *Resource Commitments: The partner is expected to provide access product development infrastructure, and technical personnel for formulation, testing, and regulatory documentation* *Collaboration Efforts: The SME will actively participate in joint project meetings, share market insights, and support the transfer of research results into commercially viable products for the Korean and international pet care market.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*This joint R&D project is expected to deliver the first, in-vitro reporter skin model for dogs, enabling reproducible species-specific cruelty-free testing of grooming product ingredients. Technological achievements include the development of stable canine reporter cell line – made donor-independent, reproducible, and available at any time through targeted cell immortalization - validated 3D skin models, and quantitative test protocols for sensitization and skin barrier integrity. These innovations will allow for the reliable identification and evaluation of both harmful and protective effects of ingredients, supporting the creation of safe and effective pet care products.* *Commercialization potential is high, as the new test system enables the rapid development and market introduction of innovative, skin-friendly grooming products tailored to the needs of dogs. The project addresses a fast-growing global market for pet skin and coat care, with significant opportunities in Korea and internationally.* *Broader industrial and economic impact includes improved animal welfare through the reduction of animal testing, enhanced product safety, and the establishment of new standards for species-specific cosmetic testing. The project will strengthen scientific and technological collaboration, expand market access, and contribute to the growth of the pet care industry.*

6. References

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